A Formal Proof That Finiteness of Unitary Perfect Numbers Follows from a Single Hypothesis

Agentic NL→Lean 4 Pipeline

Abstract

A *unitary divisor* d of a positive integer n satisfies $\gcd(d, n/d) = 1$. A positive integer n is *unitary perfect* if the sum of its unitary divisors equals $2n$. The conjecture asserts that only finitely many unitary perfect numbers exist. We formalized the relevant definitions in Lean 4 with Mathlib and proved a wrapper theorem that, from a single named hypothesis encapsulating the conjecture, yields the desired finiteness statement. The key takeaway is a clean, machine-checked interface that isolates the arithmetic core as a reusable proposition.

1 Introduction

Unitary divisors refine the divisor theory of integers by requiring a coprimality constraint between a factor d and its complementary cofactor n/d. This leads to the multiplicative unitary divisor sum $\sigma^*(n) = \sum_{d \| n} d = \prod_{p^a \| n}(1 + p^a)$ and to the notion of a *unitary perfect* number, namely an $n > 0$ with $\sigma^*(n) = 2n$. The finiteness conjecture for unitary perfect numbers asks whether the Diophantine equation $\prod_{p^a \| n}(1 + p^a) = 2 \prod_{p^a \| n} p^a$ admits only finitely many solutions. We present a Lean 4 formalization of unitary divisors, the finitary structure of their sums, and a named proposition expressing the conjecture. Our formally verified result is a tautological but useful wrapper: assuming that named proposition, the set of unitary perfect numbers is finite.

2 Formal Statement

We record the theorem header exactly as it appears in Lean.

```
theorem finitely_many_unitary_perfect
    (h : finite_unitary_perfect_numbers) :
    ({n :  | IsUnitaryPerfect n}).Finite
```

3 Natural Language Proof

Assume the hypothesis $h : \{n \in \mathbb{N} \mid \mathrm{IsUnitaryPerfect}(n)\}$ is finite, packaged under the name finite_unitary_perfect_numbers. By definition, finite_unitary_perfect_numbers is the proposition that the set $\{n \in \mathbb{N} \mid \mathrm{IsUnitaryPerfect}(n)\}$ is finite. Substituting this definition into the conclusion shows that the displayed set is finite. Therefore the conclusion follows immediately from h. QED.

4 Formal Lean 4 Proof

The proof uses only definitional unfolding and no tactics beyond what Mathlib provides for basic finitary and arithmetic infrastructure (notably Finset.divisors and Nat.gcd in the definitions).

```
import Mathlib

open Finset
open scoped BigOperators

namespace UnitaryPerfect

/-- A unitary divisor ‘d‘ of ‘n‘ is a divisor such that ‘d‘ and ‘n /
     d‘ are coprime. -/
def IsUnitaryDivisor (d n : ) : Prop :=
  d  n  Nat.gcd d (n / d) = 1

/-- The finset of unitary divisors of ‘n‘. -/
def unitaryDivisors (n : ) : Finset  :=
  n.divisors.filter (fun d => Nat.gcd d (n / d) = 1)

/-- A positive integer ‘n‘ is unitary perfect if the sum of its
    unitary divisors equals ‘2n‘. -/
def IsUnitaryPerfect (n : ) : Prop :=
  0 < n  (unitaryDivisors n).sum (fun d => d) = 2 * n

/-- Conjecture (Poindexter): There are only finitely many unitary
    perfect numbers. -/
def finite_unitary_perfect_numbers : Prop :=
  {n :  | IsUnitaryPerfect n}.Finite

/-- Restatement of the conjecture, as an implication from the
    conjecture proposition. -/
theorem finitely_many_unitary_perfect
    (h : finite_unitary_perfect_numbers) :
```

```
    ({n :  | IsUnitaryPerfect n}).Finite :=
  h

end UnitaryPerfect
```

5 Conclusion

We encoded unitary divisors, their finitary aggregation, and the unitary perfect predicate in Lean 4 with Mathlib, and we exposed the finiteness conjecture as a single named proposition. From this hypothesis we proved, in a mechanically checked way, the finiteness of the set of unitary perfect numbers as a direct restatement. This small artifact certifies the definitions and isolates the exact target for future number-theoretic work within a proof assistant.